PS 211: Exam 2 Review Sheet

Fall 2026

*Exam 2 will focus on content from Lectures 7 - 10, though may also refer to concepts from Lectures 1 - 6. Please note that this is not a comprehensive list of what may be on the exam, but is intended to help guide your studying. You may bring a single 8.5 x 11" piece of paper with hand-written notes (both sides). You will **not** need a calculator for the exam.*

Lecture 7: Normal distribution, standardization, & *z* scores

Normal distribution; deviations from normality (skew, "suspicious" distributions)
Standardization: comparing scores measured on different scales
Computing a z score; transforming a z score back to a raw score
Distribution of z scores vs. the standard normal (*z*) distribution
Percentiles; area under the curve as the proportion of scores

Lecture 8: Central Limit Theorem & standard error

Distribution of scores vs. distribution of means
Central Limit Theorem: shape, center, and spread of the distribution of means; the role of sample size
Standard error of the mean: how to compute it, what it measures
SD vs. SE: what each describes, and what happens to each as n grows
Why sample variance divides by N − 1 (biased vs. unbiased estimates)

Lecture 9: Percentiles, *z* for a sample mean & the six steps of hypothesis testing

Converting *z* scores to percentiles
How to determine if a score is "extreme"
Two kinds of z: a single score (divide by σ) vs. a sample mean (divide by SE)
The logic of every hypothesis test: assume the null, picture the null world, locate your result; test statistic = signal ÷ noise
The six steps of hypothesis testing applied to a *z* test: (1) populations, comparison distribution & assumptions; (2) hypotheses; (3) characteristics of the comparison distribution: center, spread, shape; (4) critical values; (5) test statistic; (6) decision
Reject vs. fail to reject (never "accept") the null

Lecture 10: Alpha, critical values, p values, one-tailed tests & confidence intervals

Alpha: a rule set in advance; alpha as the Type I error rate we agree to accept; consequences of α = .01 vs. .05 (Type I vs. Type II errors)
Critical region and critical values; where the critical region sits for two-tailed vs. one-tailed tests
p values: what they are and are not; one- vs. two-tailed p values; comparing p to alpha
Alpha vs. p value (areas) and critical value vs. test statistic (points)
Point estimates vs. interval estimates
Computing a 95% confidence interval for a mean (M ± z × SE) and interpreting it correctly